

Business Science: Mastery Study Plan

A PDF-only mastery curriculum for understanding, analyzing, building, improving, and scaling businesses through observation, testing, evidence, accounting, systems improvement, and disciplined execution.

Executive Summary

Input: 441 PDF files from 13 subject folders. Process: group sources by subject, map each subject to a business system function, sequence lessons from system foundations to integrated execution, and train every lesson through signal → meaning → value → execution. Output: a 24-lesson, 24-week curriculum with exercises, trackers, diagnostic templates, assessment, and a final operating-system project. Feedback loop: each week produces a measurable artifact, a diagnosis, and an execution rule that is reviewed against business evidence.

One-line framework: Business Science is evidence disciplined by accounting, directed by finance, executed through operations, pulled by marketing, protected by law, tax, and audit, and scaled through information systems.

Required Operating Flow

- Signal: the business data, behavior, financial result, operational issue, customer action, legal risk, tax issue, or market feedback being observed.
- Meaning: what the signal reveals about the business system.
- Value: why the signal matters for profit, cash flow, risk control, growth, efficiency, or survival.
- Execution: the business action taken to improve the system.

Source Control

Only PDF files from the scanned subject folders are used as lesson sources. Non-PDF materials were excluded from the curriculum source layer. Each lesson lists PDF sources only.

Full PDF Inventory Organized by Subject Folder

Subject	PDF count
Accounting Information Systems	20
Auditing	34
Book Publishing Business	44
Business Law	81
Corporate Finance	46
Cost Accounting	16
Financial Accounting	17
Financial Statement Analysis	32
Managerial Accounting	21
Marketing	20
Operations Management	26
System Science	27
Taxation	57
Total	441

Accounting Information Systems (20 PDFs)

- 01_OpenStax_Principles_Financial_Accounting_AIS_Chapter_7.pdf
- 02_Lumen_Financial_Accounting_AIS_and_Bank_Reconciliation.pdf
- 03_Springer_Digital_Dashboard_Use_by_Management_Accountants_2025.pdf
- 04_Springer_Business_Analytics_and_Management_Control_2025.pdf
- 05_MDPI_AIS_System_Competence_Information_Quality_Performance_2024.pdf
- 06_MDPI_REA_Triple_Entry_Accounting_Blockchain_2023.pdf
- 07_MDPI_Machine_Learning_for_Triple_Entry_Accounting_2025.pdf
- 08_AISel_REA_Design_Theory_Enterprise_System_Design_2016.pdf
- 09_Dunn_McCarthy_REA_Intellectual_Heritage_1997.pdf
- 10_MPRA_Continuous_Auditing_Technology_Review_2025.pdf
- 11_Informatica_AIS_Big_Data_Mining_Data_Cleaning_Automation.pdf
- 12_ABAAcademies_Business_Intelligence_Activating_AIS_2022.pdf
- 13_AsianInstitute_Data_Dashboarding_in_Accounting_Tableau_2023.pdf
- 14_XBRL_US_Financial_Reporting_with_Open_Data_Standards.pdf
- 15_XBRL_International_Understanding_XBRL_for_Software_Vendors.pdf
- 16_XBRL_National_Bank_of_Belgium_Case_Study.pdf
- 17_Springer_Hard_Rules_and_Soft_Rules_for_XBRL_2014.pdf
- 18_RPA_in_Accounting_Applied_Overview.pdf
- 19_HighRadius_Bank_Reconciliation_Applied_Overview.pdf
- 20_WorldBank_Financial_Management_Information_System_AIS_ERP_Dashboards.pdf

Auditing (34 PDFs)

- ACFE_Occupational_Fraud_Report_to_the_Nations_2024.pdf
- ACFE_Occupational_Fraud_Report_to_the_Nations_2026.pdf
- AICPA_AU-C_Sections_Currently_Effective_2026.pdf
- AICPA_AU-C_Whats_New_2026.pdf
- Auditing_Information_Systems_A_Practical_Approach_eCampusOntario.pdf
- Bayesian_Approach_to_Audit_Evidence_UvA_2026.pdf

- Bayesian_Benefits_for_Auditing_Derks_2023.pdf
- GAO_Assessing_Data_Reliability_2019.pdf
- GAO_CIGIE_Financial_Audit_Manual_Volume_1_2024.pdf
- GAO_CIGIE_Financial_Audit_Manual_Volume_2_2024.pdf
- GAO_CIGIE_Financial_Audit_Manual_Volume_3_2025.pdf
- GAO_Cybersecurity_Program_Audit_Guide_2023.pdf
- GAO_FISCAM_2024.pdf
- GAO_Fraud_Risk_Management_Framework_2015.pdf
- GAO_Government_Auditing_Standards_Yellow_Book_2024.pdf
- GAO_Green_Book_Internal_Control_2025.pdf
- IAASB_Handbook_2025_Volume_1_ISAs_ISQMs.pdf
- IAASB_Handbook_2025_Volume_2_ISA_for_LCE.pdf
- IAASB_Handbook_2025_Volume_3_Sustainability_Assurance.pdf
- IAASB_Handbook_2025_Volume_4_Assurance_Review_Related_Services.pdf
- IAASB_Handbook_2025_Volume_5_Audit_Quality_Framework.pdf
- IIA_Global_Internal_Audit_Standards_2024.pdf
- Internal_Auditing_A_Practical_Approach_eCampusOntario.pdf
- Monetary_Unit_Sampling_Adjusted_Empirical_Likelihood_2020.pdf
- NIST_SP_800-115_Security_Testing_Assessment.pdf
- NIST_SP_800-137_Continuous_Monitoring.pdf
- NIST_SP_800-30r1_Risk_Assessment.pdf
- NIST_SP_800-37r2_Risk_Management_Framework.pdf
- NIST_SP_800-53Ar5_Assessing_Security_Privacy_Controls.pdf
- NIST_SP_800-53B_Control_Baselines.pdf
- NIST_SP_800-53r5_Security_Privacy_Controls.pdf
- PCAOB_Auditing_Standards_FY_Beginning_After_2025-12-15.pdf
- Sampling_Audit_Evidence_Using_Naive_Bayes_Classifier_2024.pdf
- Sequential_Audit_Sampling_with_Statistical_Guarantees_2026.pdf

Book Publishing Business (44 PDFs)

- 01_IP_Rights/01_WIPO_Understanding_Copyright_and_Related_Rights.pdf
- 01_IP_Rights/02_WIPO_What_Is_Intellectual_Property.pdf
- 01_IP_Rights/03_WIPO_Economic_Contribution_of_Copyright_Industries.pdf
- 01_IP_Rights/04_USCO_Copyright_Basics.pdf
- 01_IP_Rights/05_USCO_Duration_of_Copyright.pdf
- 01_IP_Rights/06_USCO_Derivative_Works.pdf
- 01_IP_Rights/07_Authors_Alliance_Publication_Contracts.pdf
- 01_IP_Rights/08_Authors_Alliance_Rights_Reversion.pdf
- 01_IP_Rights/09_Authors_Alliance_Rights_Reversion_Letter_Guide.pdf
- 01_IP_Rights/10_Authors_Alliance_Understanding_Open_Access.pdf
- 01_IP_Rights/11_Authors_Alliance_Fair_Use_for_Nonfiction_Authors.pdf
- 01_IP_Rights/12_PA_Content_Superpower_AI_Licensing.pdf
- 01_IP_Rights/13_PA_Copyright_and_AI_Consultation_Response_2025.pdf
- 01_IP_Rights/14_PA_Opt_Out_Workability_Briefing.pdf
- 02_Market_Economics/01_WIPO_Global_Publishing_Industry_2022.pdf
- 02_Market_Economics/02_WIPO_Global_Publishing_Industry_Pre_Release.pdf
- 02_Market_Economics/03_FEP_European_Book_Publishing_Statistics_2024.pdf
- 02_Market_Economics/04_FEP_European_Book_Publishing_Statistics_2023.pdf
- 02_Market_Economics/05_FEP_European_Book_Publishing_Statistics_2022.pdf

- 02_Market_Economics/06_PA_Publishing_in_2025.pdf
- 02_Market_Economics/07_PA_Annual_Report_2025.pdf
- 02_Market_Economics/08_PA_Future_of_Learning.pdf
- 03_Operations_Metadata/01_EDITEUR_Identifiers_in_ONIX.pdf
- 03_Operations_Metadata/02_EDITEUR_Sales_Rights_in_ONIX.pdf
- 03_Operations_Metadata/03_EDITEUR_Publishing_Status_and_Availability.pdf
- 03_Operations_Metadata/04_EDITEUR_Pricing_in_ONIX_Part_1.pdf
- 03_Operations_Metadata/05_EDITEUR_Pricing_in_ONIX_Part_2.pdf
- 03_Operations_Metadata/06_EDITEUR_Products_in_Multiple_Markets_Part_1.pdf
- 03_Operations_Metadata/07_EDITEUR_Products_in_Multiple_Markets_Part_2.pdf
- 03_Operations_Metadata/08_EDITEUR_Describing_Ebooks_in_ONIX.pdf
- 03_Operations_Metadata/09_EDITEUR_Describing_Open_Access_Monographs_in_ONIX.pdf
- 03_Operations_Metadata/10_EDITEUR_ONIX_Books_Digital_Products.pdf
- 03_Operations_Metadata/11_EDITEUR_Metadata_for_Translated_Texts.pdf
- 03_Operations_Metadata/12_EDITEUR_Audiobook_Products_in_ONIX.pdf
- 03_Operations_Metadata/13_EDITEUR_Mythbusting_Thema.pdf
- 04_Audience_Demand/01_NEA_Reading_at_Risk.pdf
- 04_Audience_Demand/02_NEA_To_Read_or_Not_To_Read.pdf
- 04_Audience_Demand/03_NEA_2017_Survey_Public_Participation_in_Arts.pdf
- 04_Audience_Demand/04_NEA_US_Patterns_of_Arts_Participation.pdf
- 05_Open_Publishing_Business_Models/01_Springer_Nature_Future_of_Open_Access_Books.pdf
- 05_Open_Publishing_Business_Models/02_Digital_Science_State_of_Open_Monographs.pdf
- 05_Open_Publishing_Business_Models/03_Knowledge_Exchange_Roadmap_OA_Monographs.pdf
- 05_Open_Publishing_Business_Models/04_APC_Revenue_Estimates_Six_Publishers_2019_2023.pdf
- 05_Open_Publishing_Business_Models/05_Oligopoly_of_Academic_Publishers.pdf

Business Law (81 PDFs)

- 01_Core_Textbooks/advanced-business-law-and-the-legal-environment-saylor.pdf
- 01_Core_Textbooks/business-law-and-the-legal-environment-saylor.pdf
- 01_Core_Textbooks/government-regulation-and-legal-environment-of-business-saylor.pdf
- 01_Core_Textbooks/legal-and-ethical-environment-of-business-saylor.pdf
- 02_Contracts/american-contract-law-for-a-global-age-cali.pdf
- 02_Contracts/economic-analysis-contract-law-after-three-decades-posner.pdf
- 02_Contracts/economic-theory-of-contract-law.pdf
- 02_Contracts/sales-and-leases-ucc-article-2-and-2a-cali.pdf
- 03_Governance_Legal_Structure/corporate-law-and-corporate-governance-contractual-perspective.pdf
- 03_Governance_Legal_Structure/essential-role-of-organizational-law-hansmann-kraakman.pdf
- 03_Governance_Legal_Structure/g20-oecd-principles-of-corporate-governance-2023.pdf
- 03_Governance_Legal_Structure/oecd-corporate-governance-factbook-2025.pdf
- 03_Governance_Legal_Structure/oecd-guidelines-corporate-governance-state-owned-enterprises-2024.pdf
- 04_Regulation_Compliance/anti-corruption-ethics-and-compliance-handbook-for-business.pdf
- 04_Regulation_Compliance/dodd-frank-act-2010-public-law-111-203.pdf
- 04_Regulation_Compliance/doj-evaluation-of-corporate-compliance-programs-2024.pdf
- 04_Regulation_Compliance/doj-ftc-merger-guidelines-2023.pdf
- 04_Regulation_Compliance/fcpa-resource-guide-second-edition-doj-sec.pdf
- 04_Regulation_Compliance/ftc-doj-antitrust-guidelines-business-activities-affecting-workers-2025.pdf
- 04_Regulation_Compliance/ftc-doj-antitrust-guidelines-collaborations-among-competitors.pdf
- 04_Regulation_Compliance/irs-publication-3402-tax-issues-for-llcs.pdf
- 04_Regulation_Compliance/irs-publication-5868-business-structures.pdf

- 04_Regulation_Compliance/sarbanes-oxley-act-2002-public-law-107-204.pdf
- 04_Regulation_Compliance/us-sentencing-guidelines-2025-chapter-8-organizations.pdf
- 05_Scientific_Empirical_Law/complex-societies-and-the-growth-of-law.pdf
- 05_Scientific_Empirical_Law/contract-as-automaton-computational-representation-financial-agreements.pdf
- 05_Scientific_Empirical_Law/corporate-crime-and-punishment-empirical-study.pdf
- 05_Scientific_Empirical_Law/economic-analysis-of-law-shavell.pdf
- 05_Scientific_Empirical_Law/formal-methods-business-process-compliance-systematic-review.pdf
- 05_Scientific_Empirical_Law/material-contracts-corpus.pdf
- 05_Scientific_Empirical_Law/network-of-global-corporate-control.pdf
- 05_Scientific_Empirical_Law/pile-of-law-open-source-legal-dataset.pdf
- 06_Open_Source_Contracts_Modules/A1-Introduction.pdf
- 06_Open_Source_Contracts_Modules/B1-Introduction-to-the-Bases-for-Enforceable-Promises.pdf
- 06_Open_Source_Contracts_Modules/B2-Consideration.pdf
- 06_Open_Source_Contracts_Modules/B3-Promissory-Estoppel.pdf
- 06_Open_Source_Contracts_Modules/B4-Unjust-Enrichment-and-Promises-for-Benefit-Received.pdf
- 06_Open_Source_Contracts_Modules/FD1-Introduction-to-Formation-Defenses.pdf
- 06_Open_Source_Contracts_Modules/FD2-Incapacity.pdf
- 06_Open_Source_Contracts_Modules/FD3-Public-policy.pdf
- 06_Open_Source_Contracts_Modules/FD4-Mistake.pdf
- 06_Open_Source_Contracts_Modules/FD5-Misrepresentation.pdf
- 06_Open_Source_Contracts_Modules/FD6-Duress-and-Modification.pdf
- 06_Open_Source_Contracts_Modules/FD7-Unconscionability.pdf
- 06_Open_Source_Contracts_Modules/FD8-The-Statute-of-Frauds.pdf
- 06_Open_Source_Contracts_Modules/OAN1-Introduction-to-Offer-Acceptance-and-Negotiation.pdf
- 06_Open_Source_Contracts_Modules/OAN2-Offer-and-Acceptance-I-The-Offer.pdf
- 06_Open_Source_Contracts_Modules/OAN3-Offer-and-Acceptance-II-The-Acceptance.pdf
- 06_Open_Source_Contracts_Modules/OAN4-Offer-and-Acceptance-III-Unilateral-Contracts.pdf
- 06_Open_Source_Contracts_Modules/OAN5-Offer-and-Acceptance-IV-Formation-under-the-UCC.pdf
- 06_Open_Source_Contracts_Modules/OAN5A-UCC-2-207-and-Consumer-Contracts.pdf
- 06_Open_Source_Contracts_Modules/OAN6-Mutual-Assent.pdf
- 06_Open_Source_Contracts_Modules/OAN7-Indefinite-and-Illusory-Promises.pdf
- 06_Open_Source_Contracts_Modules/OAN8-Negotiation-and-Precontractual-Obligations.pdf
- 06_Open_Source_Contracts_Modules/OAN9-Options-Contracts.pdf
- 06_Open_Source_Contracts_Modules/P1-Introduction-to-Performance-and-Breach.pdf
- 06_Open_Source_Contracts_Modules/P2-Performance-I-Express-Conditions.pdf
- 06_Open_Source_Contracts_Modules/P3-Performance-II-Constructive-Conditions-and-Material-Breach.pdf
- 06_Open_Source_Contracts_Modules/P4-Performance-III-Repudiation-and-Anticipatory-Breach.pdf
- 06_Open_Source_Contracts_Modules/P5-Performance-IV-The-Perfect-Tender-Rule.pdf
- 06_Open_Source_Contracts_Modules/P6-Excuses-for-Nonperformance.pdf
- 06_Open_Source_Contracts_Modules/R1-Introduction-to-Contract-Remedies.pdf
- 06_Open_Source_Contracts_Modules/R10-Liquidated-Damages-Attys-Fees.pdf
- 06_Open_Source_Contracts_Modules/R11-Specific-Performance.pdf
- 06_Open_Source_Contracts_Modules/R2-Establishing-Expectation-Damages.pdf
- 06_Open_Source_Contracts_Modules/R3-Calculating-Expectation-Damages.pdf
- 06_Open_Source_Contracts_Modules/R4-Damages-under-the-UCC.pdf
- 06_Open_Source_Contracts_Modules/R5-Reliance-Damages.pdf
- 06_Open_Source_Contracts_Modules/R6-Mitigation.pdf
- 06_Open_Source_Contracts_Modules/R7-Consequential-and-Incidental-Damages.pdf
- 06_Open_Source_Contracts_Modules/R8-Certainty.pdf

- 06_Open_Source_Contracts_Modules/R9-Restitution-Damages.pdf
- 06_Open_Source_Contracts_Modules/T1-Introduction-to-the-Content-of-the-Parties-Obligations.pdf
- 06_Open_Source_Contracts_Modules/T2-Interpretation.pdf
- 06_Open_Source_Contracts_Modules/T3-Implied-Terms-and-Good-Faith.pdf
- 06_Open_Source_Contracts_Modules/T4-Dueling-Provisions-and-Incorporation-by-Reference.pdf
- 06_Open_Source_Contracts_Modules/T5-Warranties.pdf
- 06_Open_Source_Contracts_Modules/TP1-The-Rights-and-Obligations-of-Third-Parties.pdf
- 06_Open_Source_Contracts_Modules/TP2-Third-Party-Beneficiaries.pdf
- 06_Open_Source_Contracts_Modules/TP3-Assignment-and-Delegation.pdf
- business_law_pdf_index.pdf

Corporate Finance (46 PDFs)

- 01_Deakin_Business_Finance_Open_Textbook_2023.pdf
- 02_Touro_Corporate_Finance_Open_Textbook_2023.pdf
- 03_Schoenmaker_Schramade_Corporate_Finance_for_Long_Term_Value_2023.pdf
- 04_Damodaran_Applied_Corporate_Finance_First_Principles_Ch1_4.pdf
- 05_Damodaran_Valuation_Approaches_and_Metrics_2006.pdf
- 06_Damodaran_Equity_Risk_Premiums_2012.pdf
- 07_Damodaran_Corporate_Finance_First_Principles_Packet.pdf
- 08_Welch_Corporate_Finance_Introduction_2022.pdf
- 09_Welch_Present_Value_NPV_2022.pdf
- 10_Welch_Stock_Bond_Valuation_Anuities_Perpetuities_2022.pdf
- 11_Welch_Capital_Budgeting_Rules_2022.pdf
- 12_Welch_Uncertainty_Default_and_Risk_2022.pdf
- 13_Welch_Investor_Choice_Risk_and_Reward_2022.pdf
- 14_Welch_Benchmarked_Costs_of_Capital_2022.pdf
- 15_Welch_CAPM_2022.pdf
- 16_Welch_Capital_Budgeting_Applications_and_Pitfalls_2022.pdf
- 17_Welch_From_Financial_Statements_to_Economic_Cash_Flows_2022.pdf
- 18_Welch_Valuation_from_Comparables_and_Ratios_2022.pdf
- 19_Welch_Capital_Structure_in_a_Perfect_Market_2022.pdf
- 20_Welch_Taxes_and_Capital_Structure_2022.pdf
- 21_Welch_Imperfect_Market_Capital_Structure_2022.pdf
- 22_Welch_Equity_Payouts_Dividends_Repurchases_2022.pdf
- 23_Welch_Pro_Forma_Financial_Statements_2022.pdf
- 24_Welch_Corporate_Finance_Companion_2020.pdf
- 25_Graham_Harvey_Theory_and_Practice_of_Corporate_Finance_2001.pdf
- 26_Graham_Harvey_Theory_and_Practice_Corporate_Finance_Data_2003.pdf
- 27_Graham_Harvey_How_CFOs_Make_Capital_Budgeting_and_Structure_Decisions_2002.pdf
- 28_Myers_Capital_Structure_Puzzle_1984.pdf
- 29_Myers_Majluf_Corporate_Financing_and_Investment_Asymmetric_Information_1984.pdf
- 30_Rajan_Zingales_What_Do_We_Know_About_Capital_Structure_1995.pdf
- 31_Baker_Wurgler_Market_Timing_and_Capital_Structure_2002.pdf
- 32_Jensen_Meckling_Theory_of_the_Firm_Agency_Costs_1976.pdf
- 33_Jensen_Agency_Costs_of_Free_Cash_Flow_Corporate_Finance_Takeovers_1986.pdf
- 34_Markowitz_Portfolio_Selection_1952.pdf
- 35_Fama_French_CAPM_Theory_and_Evidence_2004.pdf
- 36_Fama_French_Five_Factor_Asset_Pricing_Model_2015.pdf
- 37_Fama_French_Cross_Section_of_Expected_Stock_Returns_1992.pdf

- 38_Jagannathan_Meier_Tarhan_Hurdle_Rates_for_Capital_Budgeting_2011.pdf
- 39_Fazzari_Hubbard_Petersen_Financing_Constraints_and_Corporate_Investment_1988.pdf
- 40_Almeida_Campello_Financial_Constraints_Asset_Tangibility_Corporate_Investment_2006.pdf
- 41_Geyer_Klingeberg_et_al_What_Do_We_Know_About_Corporate_Hedging_2018.pdf
- 42_Bodnar_Graham_Harvey_Marston_Managing_Risk_Management_2011.pdf
- 43_Giambona_Graham_Harvey_Bodnar_Theory_Practice_Corporate_Risk_Management_2018.pdf
- 44_DeAngelo_DeAngelo_Stulz_Dividend_Policy_Agency_Costs_Earned_Equity_2004.pdf
- 45_Stein_Internal_Capital_Markets_and_Competition_for_Corporate_Resources_1997.pdf
- 46_Mauboussin_Capital_Allocation_Fundamental_Responsibility_2015.pdf

Cost Accounting (16 PDFs)

- 01_OpenStax_Principles_of_Accounting_Vol2_Manual_Accounting.pdf
- 02_Saylor_Manual_Accounting_Heisinger_Hoyle.pdf
- 03_Penn_State_Financial_and_Manual_Accounting_Paff.pdf
- 04_UNG_Principles_of_Manual_Accounting_Jonick.pdf
- 05_Global_Text_Accounting_Principles_A_Business_Perspective.pdf
- 06_Public_Domain_Cost_Accounting_Principles_and_Practice_1921.pdf
- 09_APO_Manual_on_Material_Flow_Cost_Accounting_ISO14051.pdf
- 10_MDPI_MFCA_Resource_Efficiency_Manufacturing_Schmidt_Nakajima.pdf
- 11_Springer_MFCA_Multiple_Inefficiency_Factors_Recycling.pdf
- 12_Springer_Target_Costing_Preference_Dependency.pdf
- 13_Springer_TDABC_Costing_Implementation_Strategies.pdf
- 16_Springer_Management_Accounting_Operations_Shop_Floor_Review.pdf
- 17_Springer_Simulation_Based_Management_Accounting_Control.pdf
- 18_MDPI_Essential_Factors_Designing_Cost_Accounting_System.pdf
- 19_MDPI_Life_Cycle_Cost_Management_Target_Costing.pdf
- 21_IFAC_Evaluating_and_Improving_Costing_in_Organizations.pdf

Financial Accounting (17 PDFs)

- 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- 04 - Annand Marchand Dauderis - Introduction to Financial Accounting US GAAP Adaptation 2021-A.pdf
- 05 - Saylor - Financial Accounting.pdf
- 06 - Hermanson Edwards Maher - Accounting Principles A Business Perspective.pdf
- 07 - Paff - Financial and Managerial Accounting.pdf
- 08 - LibreTexts - Financial Accounting.pdf
- 10 - Arnold Kyle - Intermediate Financial Accounting Volume 1 2021-A.pdf
- 11 - Arnold Kyle - Intermediate Financial Accounting Volume 2 2021-A.pdf
- 12 - FASB - Concepts Statement No 8 As Amended.pdf
- 13 - IFRS - Conceptual Framework for Financial Reporting 2024.pdf
- 14 - Bigel - Introduction to Financial Analysis.pdf
- 15 - MSU - Financial Management for Small Businesses 2nd OER.pdf
- 16 - Bushman Smith - Transparency Financial Accounting Information and Corporate Governance.pdf
- 19 - Information Theoretic Approach for Accounting Classification.pdf
- 20 - Accounting Statement Analysis at Industry Level Compositional Approach.pdf
- 21 - REA Triple-Entry Accounting and Blockchain.pdf
- 22 - SEC - Beginners Guide to Financial Statements.pdf

Financial Statement Analysis (32 PDFs)

- 01_Bigel_Introduction_to_Financial_Analysis_Open_Touro_2022.pdf
- 02_Penman_The_Quality_of_Financial_Statements_2003.pdf

- 03_Virginia_Tech_Analysis_of_Financial_Statements_Using_Ratios_2019.pdf
- 04_MIT_OCW_Financial_Statement_Analysis_Ratio_Analysis_2004.pdf
- 05_Texas_Southern_Financial_Analysis_Ratios_Benchmarking.pdf
- 06_NYU_Ratio_Analysis_Common_Size_and_Caveats.pdf
- 07_OSU_Using_the_F_Score_in_Detecting_Fraudulent_Financial_Reporting.pdf
- 08_AICPA_Analyzing_Financial_Ratios_Practice_Aid_2006.pdf
- 09_AgEcon_Financial_Ratio_Analysis_using_ARMS_Data_2017.pdf
- 10_CARF_2024_Financial_Ratios_and_Trend_Analysis.pdf
- 11_NACUBO_Financial_Ratio_Analysis_Project_Final_Report_1996.pdf
- 12_Nissim_Penman_Ratio_Analysis_and_Equity_Valuation_Working_Paper_1999.pdf
- 13_Nissim_Penman_Ratio_Analysis_and_Equity_Valuation_From_Research_to_Practice_2001.pdf
- 14_Nissim_Penman_FSA_of_Leverage_Profitability_Price_to_Book_2003.pdf
- 15_Ou_Penman_FSA_and_Prediction_of_Stock_Returns_1989.pdf
- 16_Beneish_Detection_of_Earnings_Manipulation_1999.pdf
- 17_Piotroski_Value_Investing_Historical_Financial_Statement_Information_2000.pdf
- 18_Altman_Financial_Ratios_Discriminant_Analysis_Bankruptcy_1968.pdf
- 19_Nissim_Analysis_and_Valuation_of_Insurance_Companies_2010.pdf
- 20_Penman_What_Matters_in_Company_Valuation_2003.pdf
- 21_Linares_Mustaros_et_al_New_Financial_Ratios_CoDa_Methodology_2022.pdf
- 22_Saus_Sala_et_al_Compositional_DuPont_Analysis_2021.pdf
- 23_Carreras_Coenders_PCA_of_Financial_Statements_CoDa_2020.pdf
- 24_Coenders_Arimany_Accounting_Statement_Analysis_Industry_Level_CoDa_2023.pdf
- 25_Cui_et_al_Accounting_Choices_in_Data_Envelopment_Analysis_2025.pdf
- 26_Cui_et_al_Data_Envelopment_Analysis_and_Accounting_Measures_2025.pdf
- 27_MDPI_Corporate_Bankruptcy_Prediction_Open_Access_Book.pdf
- 28_MDPI_Bankruptcy_Prediction_Models_Based_on_Value_Measures_2021.pdf
- 29_MDPI_Compositional_Classification_of_Financial_Statement_Profiles_2022.pdf
- 30_Phan_Chang_Leveraging_Fundamental_Analysis_for_Stock_Trend_Prediction_2024.pdf
- 31_Fundamental_Analysis_in_China_Ratios_and_Stock_Prices_2019.pdf
- 32_FinAR_Bench_Financial_Statement_Analysis_Benchmark_2025.pdf

Managerial Accounting (21 PDFs)

- 01 - OpenStax - Principles of Accounting Volume 2 Managerial Accounting.pdf
- 02 - Heisinger Hoyle - Managerial Accounting.pdf
- 03 - Jonick - Principles of Managerial Accounting.pdf
- 04 - Goedl - Principles of Managerial Accounting.pdf
- 05 - Paff - Financial and Managerial Accounting.pdf
- 06 - Hermanson Edwards Ivancevich - Accounting Principles Managerial Accounting.pdf
- 07 - IFAC - Evaluating and Improving Costing in Organizations.pdf
- 08 - IMA - Conceptual Framework for Managerial Costing.pdf
- 09 - CGMA - Global Management Accounting Principles.pdf
- 10 - Kaplan - Conceptual Foundations of the Balanced Scorecard.pdf
- 11 - Kaplan Anderson - Time-Driven Activity-Based Costing.pdf
- 12 - Henttu-Aho - Rolling Forecasting in Budgetary Control Systems.pdf
- 13 - Bourne Franco-Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- 14 - Al Jasimee Blanco-Encomienda - Participative Budgeting SEM-NCA.pdf
- 15 - Eschenbach - Agent-Based Computational Economics in Management Accounting Research.pdf
- 16 - Visedsun Terdpaong - Strategy Goal Performance Mediated by Management Accounting Systems.pdf
- 17 - MAS Organizational Variables and Managerial Performance.pdf

- 18 - Quesado Silva - Activity-Based Costing and Open Innovation.pdf
- 19 - Dokulil Popesko Kadalova - Budgetary Control Process Empirical Study.pdf
- 20 - Perez-Mendez Machado-Cabezas - Management Information Systems and Corporate Performance.pdf
- 21 - Durendez Ruiz-Palomo Garcia-Perez-de-Lema Dieguez-Soto - Management Control Systems and SME Family Firm Performance.pdf

Marketing (20 PDFs)

- 01_Foundations_Purpose_and_Market_Logic/growth_competitive_strategy_3_circles.pdf
- 01_Foundations_Purpose_and_Market_Logic/intro_consumer_behaviour.pdf
- 01_Foundations_Purpose_and_Market_Logic/openstax_principles_of_marketing.pdf
- 01_Foundations_Purpose_and_Market_Logic/saylor_principles_of_marketing.pdf
- 01_Foundations_Purpose_and_Market_Logic/strategic_marketing_global_forest_industries.pdf
- 02_Market_Understanding_Data_and_Analytics/customer_churn_prediction_model.pdf
- 02_Market_Understanding_Data_and_Analytics/customer_churn_prediction_telecom.pdf
- 02_Market_Understanding_Data_and_Analytics/marketing_analytics_methods_practice.pdf
- 02_Market_Understanding_Data_and_Analytics/marketing_mix_modeling_bayesian.pdf
- 02_Market_Understanding_Data_and_Analytics/rfm_customer_segmentation.pdf
- 02_Market_Understanding_Data_and_Analytics/social_media_marketing_analytics.pdf
- 03_Value_Brand_and_Communication_Systems/b2b_customer_based_brand_equity.pdf
- 03_Value_Brand_and_Communication_Systems/dimensions_customer_based_brand_equity.pdf
- 03_Value_Brand_and_Communication_Systems/electronic_commerce_strategic_perspective.pdf
- 03_Value_Brand_and_Communication_Systems/emarketing_7e.pdf
- 03_Value_Brand_and_Communication_Systems/launch_advertising_promotion.pdf
- 04_Acquisition_Revenue_and_Governance/clv_insights_strategic_marketing.pdf
- 04_Acquisition_Revenue_and_Governance/customer_value_clv_customer_equity.pdf
- 04_Acquisition_Revenue_and_Governance/legal_aspects_marketing_sales.pdf
- 04_Acquisition_Revenue_and_Governance/power_of_selling.pdf

Operations Management (26 PDFs)

- 01_Seneca_Introduction_to_Operations_Management_Process_Capacity.pdf
- 02_eCampusOntario_Operations_Management_Textbook.pdf
- 03_KU_Introduction_to_Supply_Chain_and_Operations_Management.pdf
- 04_Saylor_Understanding_Operations_Management.pdf
- 05_Saylor_Chapter_Operations_Management.pdf
- 06_OpenStax_Introduction_to_Business_Operations_Context.pdf
- 07_VirginiaTech_Fundamentals_of_Business_Operations_Context.pdf
- 08_TUDELFT_Operations_Research_Optimization_for_Operations.pdf
- 09_Taylor_Principles_of_Scientific_Management_Excerpt.pdf
- 10_Littles_Law_50th_Anniversary_Flow_Science.pdf
- 11_arXiv_Data_Analytics_in_Operations_Management_Review.pdf
- 12_MDPI_Throughput_Management_System_Semiconductor_Fabs.pdf
- 13_Chalmers_AI_for_Throughput_Bottleneck_Analysis.pdf
- 14_DiVA_Digital_Twin_Bottleneck_Diagnosis_and_Throughput_Improvement.pdf
- 15_Springer_Bottleneck_Identification_Throughput_Serial_Lines_2026.pdf
- 16_Springer_Capacity_Planning_Scheduling_Operating_Rooms_2025.pdf
- 17_Springer_Optimal_Configuration_Production_Planning_Control.pdf
- 18_Springer_Risk_Treatment_Energy_Oriented_Production_Planning_Control.pdf
- 19_JIEM_Strategic_Bottleneck_Identification_Make_to_Order_DBR.pdf
- 20_JIEM_Theory_of_Constraints_Make_to_Order_Case_Study.pdf
- 21_EPA_Lean_and_Clean_Value_Stream_Mapping.pdf

- 22_EPA_Environmental_Guide_to_Lean_and_Six_Sigma.pdf
- 23_Lean_Enterprise_Institute_Understanding_Lean_Thinking.pdf
- 24_MDPI_I3oT_Sub_Bottleneck_Detection_Continuous_Improvement.pdf
- 25_Springer_Stabilized_Cycle_Capacitated_Production_Planning.pdf
- Operations_Management_PDF_Index.pdf

System Science (27 PDFs)

- 01_Foundations_and_Complex_Systems/complex_systems_science_applications.pdf
- 01_Foundations_and_Complex_Systems/evolutionary_systems_thinking.pdf
- 01_Foundations_and_Complex_Systems/introduction_to_cybernetics.pdf
- 01_Foundations_and_Complex_Systems/networks_crowds_markets.pdf
- 01_Foundations_and_Complex_Systems/think_complexity.pdf
- 02_Dynamics_Feedback_and_Leverage/meadows_leverage_points.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_intro_sd_assignment_1.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_intro_sd_recitation_1.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_intro_sd_recitation_2.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_behavior.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_building.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_checklist.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_credit_card_model.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_dynamic.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_economics.pdf
- 02_Dynamics_Feedback_and_Leverage/mit_ocw_sd_feedback.pdf
- 03_System_Design_and_Engineering/nasa_se_expanded_guidance_vol1.pdf
- 03_System_Design_and_Engineering/nasa_se_expanded_guidance_vol2.pdf
- 03_System_Design_and_Engineering/nasa_systems_engineering_handbook.pdf
- 04_Business_Optimization_and_Decision_Science/algorithms_for_decision_making.pdf
- 04_Business_Optimization_and_Decision_Science/algorithms_for_optimization.pdf
- 04_Business_Optimization_and_Decision_Science/business_process_management_survey.pdf
- 04_Business_Optimization_and_Decision_Science/convex_optimization.pdf
- 04_Business_Optimization_and_Decision_Science/openstax_principles_management.pdf
- 04_Business_Optimization_and_Decision_Science/process_mining_manifesto.pdf
- 04_Business_Optimization_and_Decision_Science/wealth_of_nations_complexity_science_economics.pdf
- SYSTEM SCIENCE INDEX.pdf

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- ADB_Improving_Tax_Compliance.pdf
- IFS_Mirrlees_Dimensions_of_Tax_Design.pdf
- IFS_Mirrlees_Tax_by_Design.pdf
- IMF_Compliance_Risk_Management_for_Tax_Administration_Leaders.pdf
- IMF_Tax_Policy_Handbook.pdf
- IMF_Tax_Policy_and_Inclusive_Growth_2020.pdf
- IRS_Form_8832_Entity_Classification_Election.pdf
- IRS_Instructions_Form_1040_Schedule_C.pdf
- IRS_Instructions_Form_1065_Partnership_Return.pdf
- IRS_Instructions_Form_1118_Foreign_Tax_Credit_Corporations.pdf
- IRS_Instructions_Form_1120S_S_Corporation_Return.pdf
- IRS_Instructions_Form_1120_Corporation_Return.pdf
- IRS_Instructions_Form_2553_S_Corporation_Election.pdf

- IRS_Instructions_Form_5471_Foreign_Corporations.pdf
- IRS_Instructions_Form_8858_Foreign_Disregarded_Entities.pdf
- IRS_Instructions_Form_8865_Foreign_Partnerships.pdf
- IRS_Pub_15B_Employers_Tax_Guide_to_Fringe_Benefits.pdf
- IRS_Pub_15_Employers_Tax_Guide.pdf
- IRS_Pub_1_Your_Rights_as_a_Taxpayer.pdf
- IRS_Pub_334_Tax_Guide_for_Small_Business.pdf
- IRS_Pub_3402_Taxation_of_LLCs.pdf
- IRS_Pub_3498_The_Examination_Process.pdf
- IRS_Pub_463_Travel_Gift_and_Car_Expenses.pdf
- IRS_Pub_505_Withholding_and_Estimated_Tax.pdf
- IRS_Pub_515_Withholding_on_Nonresident_Aliens_and_Foreign_Entities.pdf
- IRS_Pub_535_Business_Expenses_Archived_2022.pdf
- IRS_Pub_537_Installment_Sales.pdf
- IRS_Pub_538_Accounting_Periods_and_Methods.pdf
- IRS_Pub_541_Partnerships.pdf
- IRS_Pub_542_Corporations.pdf
- IRS_Pub_544_Sales_and_Other_Dispositions_of_Assets.pdf
- IRS_Pub_551_Basis_of_Assets.pdf
- IRS_Pub_556_Examination_of>Returns_Appeal_Rights_and_Refund_Claims.pdf
- IRS_Pub_560_Retirement_Plans_for_Small_Business.pdf
- IRS_Pub_583_Starting_a_Business_and_Keeping_Records.pdf
- IRS_Pub_587_Business_Use_of_Your_Home.pdf
- IRS_Pub_594_IRS_Collection_Process.pdf
- IRS_Pub_925_Passive_Activity_and_At-Risk_Rules.pdf
- IRS_Pub_946_How_to_Depreciate_Property.pdf
- IRS_Tax_Evasion_at_the_Top_of_the_Income_Distribution.pdf
- IRS_Tax_Gap_Methodology_Publication_6031.pdf
- IRS_Tax_Gap_Projections_Publication_5869.pdf
- JCT_Estimates_of_Federal_Tax_Expenditures_2025_2029.pdf
- JCT_Overview_of_Federal_Tax_System_2025.pdf
- NBER_Behavioralist_as_Tax_Collector_2014.pdf
- NBER_General_Model_of_Behavioral_Response_to_Taxation_1998.pdf
- NBER_Tax_Avoidance_Evasion_and_Administration_Slemrod_Yitzhaki.pdf
- NBER_Tax_Compliance_and_Enforcement_Slemrod_2018.pdf
- OECD_BEPS_Transfer_Pricing_Financial_Transactions_2020.pdf
- OECD_Compliance_Risk_Management_Tax_Compliance.pdf
- OECD_Corporate_Tax_Statistics_2025.pdf
- OECD_Global_Minimum_Tax_Implementation_Toolkit_2026.pdf
- OECD_Model_Tax_Convention_Condensed_2017.pdf
- OECD_Tax_Administration_2024.pdf
- Treasury_Greenbook_FY2025_General_Explanations.pdf
- World_Bank_Behavioral_Insights_for_Tax_Compliance_2025.pdf
- arXiv_TaxSolver_Framework_for_AI_Tax_Policy_Design_2025.pdf

Curriculum Map

Phase	System Function	Lessons	Focus
1	Business as a System	Business Science Operating Logic; Business Model as Value Conversion Engine	system logic, value creation, constraints, feedback, profit logic
2	Financial Accounting and Business Measurement	Transaction Flow and Financial Statement Architecture; Accounts as Signals of Business Behavior	transaction flow, accounts, statements, performance signals
3	Financial Statement Analysis	Ratio Diagnostics for Business Health; Earnings Quality, Fraud Signals, and Failure Risk	ratios, cash flow, quality, diagnosis, failure risk
4	Managerial and Cost Accounting	Cost Behavior, Margins, and Pricing Floors; Budgeting, Variance, and Internal Performance Control	cost behavior, margins, budgeting, variance, internal control
5	Corporate Finance and Capital Allocation	Time Value, NPV, and Investment Decisions; Risk, Capital Structure, and Long-Term Value	NPV, risk, capital structure, investment, long-term value
6	Operations and Process Improvement	Flow, Throughput, and Bottleneck Diagnosis; Lean Quality, Capacity, and Scalable Execution	workflow, bottlenecks, throughput, quality, scalable execution
7	Marketing and Demand Creation	Customer Behavior, Positioning, and Offer Design; Funnel Analytics, CLV, Churn, and Demand Feedback	customer behavior, positioning, funnels, CLV, retention
8	Law, Taxation, and Risk Control	Business Structure, Contracts, and Liability; Tax Design, Compliance, and Cash Risk	entity structure, contracts, tax, compliance, liability
9	Auditing and Control Systems	Internal Controls, Evidence, and Fraud Prevention; Data Reliability, Cyber Controls, and Audit Analytics	controls, evidence, fraud prevention, data reliability
10	Accounting Information Systems	Transaction Data Architecture and Reporting Standards; Dashboards, Automation, and Decision Support	data flow, standards, dashboards, automation, decision support
11	Book Publishing Business as Applied Business Science	Publishing Product System: IP, Metadata, and Distribution; Publishing Profit System: Audience, Pricing, and Open Models	IP, metadata, audience, pricing, title economics
12	Integrated Business Science Mastery	Full Business Diagnosis: From Statements to System Levers; Build the Business Science Operating System	cross-functional diagnosis and operating-system design

Individual Lesson Plans

Lesson 1: Business Science Operating Logic

Phase	Phase 1: Business as a System
Main business science skill	Map a business as an evidence-driven money system.
Subjects used	System Science, Financial Accounting, Operations Management
Business system being studied	The complete business operating loop from resources to customer value to financial result.
Measurable outcome	Build one business system map using signal → meaning → value → execution.

PDF Sources Used

- System Science / 01_Foundations_and_Complex_Systems/introduction_to_cybernetics.pdf
- System Science / 02_Dynamics_Feedback_and_Leverage/meadows_leverage_points.pdf
- Financial Accounting / 22 - SEC - Beginners Guide to Financial Statements.pdf
- Operations Management / 01_Seneca_Introduction_to_Operations_Management_Process_Capacity.pdf

Core Concepts

- business boundary
- inputs, processes, outputs
- feedback and control
- capacity and constraints
- financial statements as system readouts
- profit as tested value creation

signal → meaning → value → execution Drill

Signal	Revenue grows while cash balance declines.
Meaning	The sales system may be outpacing collection, fulfillment, or margin control.
Value	Cash survival matters more than reported activity.
Execution	Map receivables, inventory, capacity, and payment timing before increasing sales effort.

Practical Exercise

Build one one-page business system map for any real business with five signals: demand, throughput, revenue, gross margin, and cash.

Real-World Application

Use the map to explain why a business with high sales can still fail if capacity, margin, or cash conversion breaks.

Reflection Questions

1. What business signal would you observe first when studying Business Science Operating Logic?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of business boundary in this lesson?
2. How does inputs, processes, outputs change the way you interpret business performance?

3. Why does this lesson treat the business as the complete business operating loop from resources to customer value to financial result.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Map a business as an evidence-driven money system to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 2: Business Model as Value Conversion Engine

Phase	Phase 1: Business as a System
Main business science skill	Translate strategy into a repeatable value creation model.
Subjects used	System Science, Marketing, Corporate Finance, Managerial Accounting
Business system being studied	The value conversion system that turns customer problems, capabilities, and capital into durable profit.
Measurable outcome	Create one business model map with assumptions, signals, and decision rules.

PDF Sources Used

- System Science / 04_Business_Optimization_and_Decision_Science/openstax_principles_management.pdf
- Marketing / 01_Foundations_Purpose_and_Market_Logic/growth_competitive_strategy_3_circles.pdf
- Corporate Finance / 03_Schoenmaker_Schramade_Corporate_Finance_for_Long_Term_Value_2023.pdf
- Managerial Accounting / 09 - CGMA - Global Management Accounting Principles.pdf

Core Concepts

- customer value proposition
- competitive advantage
- management accounting as decision support
- long-term value
- strategic feedback
- business model assumptions

signal → meaning → value → execution Drill

Signal	Customers buy once but do not repurchase.
Meaning	The offer may solve an acquisition problem but not a retention problem.
Value	Weak retention raises marketing cost and lowers lifetime value.
Execution	Redesign the offer, onboarding, or post-sale system and track repeat purchase rate.

Practical Exercise

Choose one business and write its value logic in four equations: customer problem, offer, operating capability, and capital return.

Real-World Application

Use the equations to compare two business models and identify which one has stronger long-term value logic.

Reflection Questions

1. What business signal would you observe first when studying Business Model as Value Conversion Engine?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of customer value proposition in this lesson?
2. How does competitive advantage change the way you interpret business performance?
3. Why does this lesson treat the business as the value conversion system that turns customer problems, capabilities, and capital into durable profit.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Translate strategy into a repeatable value creation model to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 3: Transaction Flow and Financial Statement Architecture

Phase	Phase 2: Financial Accounting and Business Measurement
Main business science skill	Trace transactions into financial statements and business meaning.
Subjects used	Financial Accounting, Accounting Information Systems, Business Law
Business system being studied	The accounting measurement system that converts business events into reliable financial signals.
Measurable outcome	Trace ten transactions through one accounting system map.

PDF Sources Used

- Financial Accounting / 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- Financial Accounting / 12 - FASB - Concepts Statement No 8 As Amended.pdf
- Accounting Information Systems / 01_OpenStax_Principles_Financial_Accounting_AIS_Chapter_7.pdf
- Business Law / 01_Core_Textbooks/business-law-and-the-legal-environment-saylor.pdf

Core Concepts

- transaction recognition
- journal to ledger flow
- assets, liabilities, equity
- income statement, balance sheet, cash flow statement
- source documents
- legal obligations behind accounting events

signal → meaning → value → execution Drill

Signal	A sale is recorded before the customer obligation is complete.
Meaning	Revenue recognition may be ahead of economic performance.
Value	Overstated revenue distorts profitability, taxes, financing decisions, and trust.
Execution	Reconcile contract terms, delivery evidence, invoice timing, and revenue recognition policy.

Practical Exercise

Trace ten common transactions from source document to journal entry to financial statement impact.

Real-World Application

Diagnose whether a revenue event, expense event, financing event, or legal obligation is being measured correctly.

Reflection Questions

1. What business signal would you observe first when studying Transaction Flow and Financial Statement Architecture?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of transaction recognition in this lesson?
2. How does journal to ledger flow change the way you interpret business performance?
3. Why does this lesson treat the business as the accounting measurement system that converts business events into reliable financial signals.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?

2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Trace transactions into financial statements and business meaning to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 4: Accounts as Signals of Business Behavior

Phase	Phase 2: Financial Accounting and Business Measurement
Main business science skill	Read accounting categories as evidence of business activity.
Subjects used	Financial Accounting, Marketing, Operations Management
Business system being studied	The business measurement layer that connects market behavior, operations, and financial accounts.
Measurable outcome	Create one account-signal dictionary for a target business.

PDF Sources Used

- Financial Accounting / 05 - Saylor - Financial Accounting.pdf
- Financial Accounting / 13 - IFRS - Conceptual Framework for Financial Reporting 2024.pdf
- Marketing / 01_Foundations_Purpose_and_Market_Logic/openstax_principles_of_marketing.pdf
- Operations Management / 06_OpenStax_Introduction_to_Business_Operations_Context.pdf

Core Concepts

- revenue as demand evidence
- expenses as resource consumption
- assets as controlled capacity
- liabilities as claims and constraints
- equity as residual ownership
- matching business activity to accounting category

signal → meaning → value → execution Drill

Signal	Inventory rises faster than revenue.
Meaning	The business may be buying or producing ahead of demand.
Value	Cash, storage cost, obsolescence, and margin risk may be increasing.
Execution	Compare inventory days, sales forecast accuracy, supplier terms, and markdown exposure.

Practical Exercise

Classify 30 business events into accounts, then write the operating behavior each account is signaling.

Real-World Application

Use the account map to separate sales growth from profitable growth and activity from value creation.

Reflection Questions

1. What business signal would you observe first when studying Accounts as Signals of Business Behavior?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of revenue as demand evidence in this lesson?
2. How does expenses as resource consumption change the way you interpret business performance?
3. Why does this lesson treat the business as the business measurement layer that connects market behavior, operations, and financial accounts.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Read accounting categories as evidence of business activity to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 5: Ratio Diagnostics for Business Health

Phase	Phase 3: Financial Statement Analysis
Main business science skill	Use ratios to diagnose liquidity, solvency, profitability, and efficiency.
Subjects used	Financial Statement Analysis, Corporate Finance, Managerial Accounting
Business system being studied	The financial diagnosis system that translates accounting outputs into business health judgments.
Measurable outcome	Calculate five financial ratios and convert each into one business action hypothesis.

PDF Sources Used

- Financial Statement Analysis / 01_Bigel_Introduction_to_Financial_Analysis_Open_Touro_2022.pdf
- Financial Statement Analysis / 03_Virginia_Tech_Analysis_of_Financial_Statements_Using_Ratios_2019.pdf
- Corporate Finance / 17_Welch_From_Financial_Statements_to_Economic_Cash_Flows_2022.pdf
- Managerial Accounting / 10 - Kaplan - Conceptual Foundations of the Balanced Scorecard.pdf

Core Concepts

- liquidity ratios
- solvency ratios
- profitability ratios
- efficiency ratios
- cash flow conversion
- balanced performance measurement

signal → meaning → value → execution Drill

Signal	Gross margin improves but operating margin declines.
Meaning	Direct unit economics may be improving while overhead or growth cost is expanding faster.
Value	Profit quality depends on the full operating structure, not just product margin.
Execution	Analyze contribution margin, fixed cost growth, headcount, acquisition cost, and operating leverage.

Practical Exercise

Calculate at least five ratios from one company's financial statements and write one operational hypothesis for each ratio.

Real-World Application

Use ratio patterns to identify whether the bottleneck is demand, margin, asset use, debt load, or cash conversion.

Reflection Questions

1. What business signal would you observe first when studying Ratio Diagnostics for Business Health?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of liquidity ratios in this lesson?
2. How does solvency ratios change the way you interpret business performance?
3. Why does this lesson treat the business as the financial diagnosis system that translates accounting outputs into business health judgments.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Use ratios to diagnose liquidity, solvency, profitability, and efficiency to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 6: Earnings Quality, Fraud Signals, and Failure Risk

Phase	Phase 3: Financial Statement Analysis
Main business science skill	Detect unreliable performance signals and financial distress patterns.
Subjects used	Financial Statement Analysis, Auditing, Accounting Information Systems
Business system being studied	The reliability system that separates real performance from distorted measurement.
Measurable outcome	Build one financial risk red-flag checklist.

PDF Sources Used

- Financial Statement Analysis / 02_Penman_The_Quality_of_Financial_Statements_2003.pdf
- Financial Statement Analysis / 16_Beneish_Detection_of_Earnings_Manipulation_1999.pdf
- Financial Statement Analysis / 18_Altman_Financial_Ratios_Discriminant_Analysis_Bankruptcy_1968.pdf
- Auditing / ACFE_Occupational_Fraud_Report_to_the_Nations_2026.pdf
- Auditing / GAO_Assessing_Data_Reliability_2019.pdf

Core Concepts

- earnings quality
- manipulation indicators
- bankruptcy risk
- fraud risk factors
- data reliability
- audit evidence as signal validation

signal → meaning → value → execution Drill

Signal	Receivables grow much faster than revenue.
Meaning	Revenue quality, collection discipline, or customer credit risk may be weakening.
Value	Reported profit may not convert into cash.
Execution	Review days sales outstanding, allowance policy, customer concentration, and invoice aging.

Practical Exercise

Create a red-flag matrix with financial, behavioral, and data-quality indicators for one business.

Real-World Application

Use the matrix to decide whether a business should be trusted, investigated, remeasured, or avoided.

Reflection Questions

1. What business signal would you observe first when studying Earnings Quality, Fraud Signals, and Failure Risk?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of earnings quality in this lesson?
2. How does manipulation indicators change the way you interpret business performance?
3. Why does this lesson treat the business as the reliability system that separates real performance from distorted measurement.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Detect unreliable performance signals and financial distress patterns to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 7: Cost Behavior, Margins, and Pricing Floors

Phase	Phase 4: Managerial and Cost Accounting
Main business science skill	Model how costs behave as volume, mix, and process choices change.
Subjects used	Managerial Accounting, Cost Accounting, Marketing
Business system being studied	The margin system that connects product design, pricing, cost structure, and profit.
Measurable outcome	Create one cost behavior and pricing model.

PDF Sources Used

- Managerial Accounting / 01 - OpenStax - Principles of Accounting Volume 2 Managerial Accounting.pdf
- Cost Accounting / 21_IFAC_Evaluating_and_Improving_Costing_in_Organizations.pdf
- Cost Accounting / 12_Springer_Target_Costing_Preference_Dependency.pdf
- Marketing / 01_Foundations_Purpose_and_Market_Logic/saylor_principles_of_marketing.pdf

Core Concepts

- fixed, variable, and mixed costs
- contribution margin
- break-even analysis
- target costing
- price floor
- customer value and willingness to pay

signal → meaning → value → execution Drill

Signal	Unit sales rise after a discount but total contribution margin falls.
Meaning	Volume is not compensating for price and margin loss.
Value	Demand growth without contribution growth destroys profit capacity.
Execution	Recalculate break-even volume, segment discounting, and offer design before repeating the promotion.

Practical Exercise

Build a cost behavior model with fixed cost, variable cost, price, contribution margin, break-even units, and target profit units.

Real-World Application

Use the model to test whether a price discount, product bundle, or volume goal improves profit.

Reflection Questions

1. What business signal would you observe first when studying Cost Behavior, Margins, and Pricing Floors?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of fixed, variable, and mixed costs in this lesson?
2. How does contribution margin change the way you interpret business performance?
3. Why does this lesson treat the business as the margin system that connects product design, pricing, cost structure, and profit?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Model how costs behave as volume, mix, and process choices change to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 8: Budgeting, Variance, and Internal Performance Control

Phase	Phase 4: Managerial and Cost Accounting
Main business science skill	Use budgets and variances to control execution without hiding reality.
Subjects used	Managerial Accounting, Cost Accounting, Operations Management
Business system being studied	The management control system that converts plans into monitored execution and corrective action.
Measurable outcome	Build one budget variance analysis and corrective action log.

PDF Sources Used

- Managerial Accounting / 02 - Heisinger Hoyle - Managerial Accounting.pdf
- Managerial Accounting / 12 - Henttu-Aho - Rolling Forecasting in Budgetary Control Systems.pdf
- Managerial Accounting / 19 - Dokulil Popesko Kadalova - Budgetary Control Process Empirical Study.pdf
- Operations Management / 08_TU Delft_Operations_Research_Optimization_for_Operations.pdf

Core Concepts

- budget as planning model
- variance analysis
- rolling forecast
- controllable and uncontrollable variance
- optimization tradeoffs
- performance accountability

signal → meaning → value → execution Drill

Signal	Labor cost variance is unfavorable while throughput is flat.
Meaning	The system may be using more labor without producing more value.
Value	Execution efficiency and margin are deteriorating.
Execution	Review staffing schedule, rework, queue time, training, and process constraints.

Practical Exercise

Create a simple flexible budget and variance report for revenue, labor, material, overhead, and cash.

Real-World Application

Use variances to decide whether a miss came from price, volume, efficiency, capacity, or planning error.

Reflection Questions

1. What business signal would you observe first when studying Budgeting, Variance, and Internal Performance Control?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of budget as planning model in this lesson?
2. How does variance analysis change the way you interpret business performance?
3. Why does this lesson treat the business as the management control system that converts plans into monitored execution and corrective action.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Use budgets and variances to control execution without hiding reality to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 9: Time Value, NPV, and Investment Decisions

Phase	Phase 5: Corporate Finance and Capital Allocation
Main business science skill	Evaluate whether capital should be committed to a business action.
Subjects used	Corporate Finance, Financial Statement Analysis, Managerial Accounting
Business system being studied	The capital allocation system that turns forecasted cash consequences into go/no-go decisions.
Measurable outcome	Build one capital budgeting model with sensitivity analysis.

PDF Sources Used

- Corporate Finance / 09_Welch_Present_Value_NPV_2022.pdf
- Corporate Finance / 11_Welch_Capital_Budgeting_Rules_2022.pdf
- Corporate Finance / 16_Welch_Capital_Budgeting_Applications_and_Pitfalls_2022.pdf
- Corporate Finance / 04_Damodaran_Applied_Corporate_Finance_First_Principles_Ch1_4.pdf
- Financial Statement Analysis / 20_Penman_What_Matters_in_Company_Valuation_2003.pdf

Core Concepts

- time value of money
- net present value
- capital budgeting

- reinvestment
- free cash flow
- decision pitfalls

signal → meaning → value → execution Drill

Signal	A project has positive accounting income but negative NPV.
Meaning	Reported profit is not compensating for capital cost, timing, or risk.
Value	Growth that earns below the cost of capital destroys long-term value.
Execution	Change price, cost, utilization, investment size, or reject the project.

Practical Exercise

Build one NPV model for a project with initial investment, operating cash flow, terminal value, discount rate, and sensitivity table.

Real-World Application

Use the model to reject a project that grows revenue but fails to earn enough return on capital.

Reflection Questions

1. What business signal would you observe first when studying Time Value, NPV, and Investment Decisions?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of time value of money in this lesson?
2. How does net present value change the way you interpret business performance?
3. Why does this lesson treat the business as the capital allocation system that turns forecasted cash consequences into go/no-go decisions.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Evaluate whether capital should be committed to a business action to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 10: Risk, Capital Structure, and Long-Term Value

Phase	Phase 5: Corporate Finance and Capital Allocation
Main business science skill	Design financing choices that support resilience and return.
Subjects used	Corporate Finance, Business Law, Taxation
Business system being studied	The financing system that balances control, flexibility, tax effects, and survival risk.
Measurable outcome	Draft one capital allocation and financing decision memo.

PDF Sources Used

- Corporate Finance / 12_Welch_Uncertainty_Default_and_Risk_2022.pdf
- Corporate Finance / 20_Welch_Taxes_and_Capital_Structure_2022.pdf
- Corporate Finance / 28_Myers_Capital_Structure_Puzzle_1984.pdf
- Corporate Finance / 46_Mauboussin_Capital_Allocation_Fundamental_Responsibility_2015.pdf
- Business Law / 03_Governance_Legal_Structure/g20-oecd-principles-of-corporate-governance-2023.pdf
- Taxation / JCT_Overview_of_Federal_Tax_System_2025.pdf

Core Concepts

- risk and return
- debt capacity
- tax shield
- agency cost
- governance
- capital allocation discipline

signal → meaning → value → execution Drill

Signal	Return on equity rises after adding debt while cash coverage weakens.
Meaning	Leverage is amplifying returns and fragility at the same time.
Value	Higher equity returns are not valuable if default risk or strategic inflexibility rises too much.
Execution	Stress test interest coverage, covenant headroom, tax benefit, and downside cash flows.

Practical Exercise

Create a capital structure memo comparing debt, equity, retained earnings, and hybrid financing for one business.

Real-World Application

Use the memo to decide how much financial risk the business can carry without weakening execution.

Reflection Questions

1. What business signal would you observe first when studying Risk, Capital Structure, and Long-Term Value?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of risk and return in this lesson?
2. How does debt capacity change the way you interpret business performance?
3. Why does this lesson treat the business as the financing system that balances control, flexibility, tax effects, and survival risk.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design financing choices that support resilience and return to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 11: Flow, Throughput, and Bottleneck Diagnosis

Phase	Phase 6: Operations and Process Improvement
Main business science skill	Find the constraint that limits business output and cash conversion.
Subjects used	Operations Management, Managerial Accounting, System Science
Business system being studied	The execution system that converts work into delivered value under capacity constraints.
Measurable outcome	Identify one operational bottleneck and one throughput improvement action.

PDF Sources Used

- Operations Management / 02_eCampusOntario_Operations_Management_Textbook.pdf
- Operations Management / 10_Littles_Law_50th_Anniversary_Flow_Science.pdf
- Operations Management / 20_JIEM_Theory_of_Constraints_Make_to_Order_Case_Study.pdf
- Managerial Accounting / 11 - Kaplan Anderson - Time-Driven Activity-Based Costing.pdf
- System Science / 04_Business_Optimization_and_Decision_Science/business_process_management_survey.pdf

Core Concepts

- process flow
- throughput
- work in process
- cycle time
- constraint
- time-driven activity-based costing

signal → meaning → value → execution Drill

Signal	Work piles up before one process step while downstream teams wait.
Meaning	The constrained resource determines system throughput.
Value	Revenue, delivery speed, labor efficiency, and customer experience all depend on the bottleneck.
Execution	Exploit the constraint, reduce setup time, shift work, and protect bottleneck time from low-value tasks.

Practical Exercise

Map one process, measure cycle time at each step, identify the bottleneck, and estimate the cost of waiting.

Real-World Application

Use the bottleneck to decide where not to spend effort: improving nonconstraints rarely improves total output.

Reflection Questions

1. What business signal would you observe first when studying Flow, Throughput, and Bottleneck Diagnosis?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of process flow in this lesson?
2. How does throughput change the way you interpret business performance?
3. Why does this lesson treat the business as the execution system that converts work into delivered value under capacity constraints.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Find the constraint that limits business output and cash conversion to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 12: Lean Quality, Capacity, and Scalable Execution

Phase	Phase 6: Operations and Process Improvement
Main business science skill	Improve workflow quality and capacity without adding unnecessary complexity.
Subjects used	Operations Management, Cost Accounting, System Science
Business system being studied	The continuous improvement system that removes waste, stabilizes quality, and increases scalable capacity.
Measurable outcome	Create one operations improvement plan with before-and-after metrics.

PDF Sources Used

- Operations Management / 21_EPA_Lean_and_Clean_Value_Stream_Mapping.pdf
- Operations Management / 23_Lean_Enterprise_Institute_Understanding_Lean_Thinking.pdf
- Operations Management / 11_arXiv_Data_Analytics_in_Operations_Management_Review.pdf
- Cost Accounting / 09_APO_Manual_on_Material_Flow_Cost_Accounting_ISO14051.pdf
- System Science / 04_Business_Optimization_and_Decision_Science/process_mining_manifesto.pdf

Core Concepts

- value stream mapping
- waste
- quality at source
- capacity planning
- material flow cost
- process mining

signal → meaning → value → execution Drill

Signal	Rework consumes capacity but is not visible in standard cost reports.
Meaning	The system hides quality failure inside normal labor and material usage.
Value	Hidden rework reduces throughput, margin, and reliability.
Execution	Track defects, rework hours, scrap, queue time, and root causes in the process dashboard.

Practical Exercise

Create a current-state and future-state value stream map for one revenue-producing workflow.

Real-World Application

Use the future-state map to reduce delay, rework, defect cost, and customer waiting time.

Reflection Questions

1. What business signal would you observe first when studying Lean Quality, Capacity, and Scalable Execution?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of value stream mapping in this lesson?
2. How does waste change the way you interpret business performance?
3. Why does this lesson treat the business as the continuous improvement system that removes waste, stabilizes quality, and increases scalable capacity.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Improve workflow quality and capacity without adding unnecessary complexity to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 13: Customer Behavior, Positioning, and Offer Design

Phase	Phase 7: Marketing and Demand Creation
Main business science skill	Design demand systems around customer behavior and differentiated value.
Subjects used	Marketing, Managerial Accounting, Corporate Finance
Business system being studied	The demand creation system that turns customer insight into offers, channels, conversion, and retained value.
Measurable outcome	Create one offer and positioning model for a target customer segment.

PDF Sources Used

- Marketing / 01_Foundations_Purpose_and_Market_Logic/openstax_principles_of_marketing.pdf
- Marketing / 01_Foundations_Purpose_and_Market_Logic/intro_consumer_behaviour.pdf
- Marketing / 01_Foundations_Purpose_and_Market_Logic/growth_competitive_strategy_3_circles.pdf
- Managerial Accounting / 13 - Bourne Franco-Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- Corporate Finance / 03_Schoenmaker_Schramade_Corporate_Finance_for_Long_Term_Value_2023.pdf

Core Concepts

- customer problem
- segmentation
- positioning
- offer design
- brand promise
- demand metrics

signal → meaning → value → execution Drill

Signal	Traffic increases but conversion rate falls.
Meaning	The business may be attracting weaker-fit customers or misaligning message and offer.
Value	Demand quality matters more than raw attention.
Execution	Segment traffic, compare channel intent, test message-market fit, and adjust offer proof.

Practical Exercise

Build one offer map with target segment, problem, promise, proof, price, channel, and expected conversion signal.

Real-World Application

Use the map to redesign an offer that attracts attention but fails to convert.

Reflection Questions

1. What business signal would you observe first when studying Customer Behavior, Positioning, and Offer Design?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of customer problem in this lesson?
2. How does segmentation change the way you interpret business performance?
3. Why does this lesson treat the business as the demand creation system that turns customer insight into offers, channels, conversion, and retained value.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design demand systems around customer behavior and differentiated value to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 14: Funnel Analytics, CLV, Churn, and Demand Feedback

Phase	Phase 7: Marketing and Demand Creation
Main business science skill	Convert market behavior into measurable acquisition and retention decisions.
Subjects used	Marketing, Accounting Information Systems, Financial Statement Analysis
Business system being studied	The customer economics system that links acquisition, conversion, retention, and profit.
Measurable outcome	Analyze one marketing funnel and calculate customer lifetime value.

PDF Sources Used

- Marketing / 02_Market_Understanding_Data_and_Analytics/marketing_analytics_methods_practice.pdf
- Marketing / 04_Acquisition_Revenue_and_Governance/customer_value_clv_customer_equity.pdf
- Marketing / 02_Market_Understanding_Data_and_Analytics/customer_churn_prediction_model.pdf
- Marketing / 02_Market_Understanding_Data_and_Analytics/rfm_customer_segmentation.pdf
- Accounting Information Systems / 03_Springer_Digital_Dashboard_Use_by_Management_Accountants_2025.pdf
- Financial Statement Analysis / 08_AICPA_Analyzing_Financial_Ratios_Practice_Aid_2006.pdf

Core Concepts

- funnel conversion
- customer lifetime value
- churn
- RFM segmentation
- dashboard design
- marketing-finance feedback

signal → meaning → value → execution Drill

Signal	A channel has low acquisition cost but high churn.
Meaning	The channel may produce low-fit customers or poor expectations.
Value	Cheap acquisition is expensive when retention and margin are weak.
Execution	Compare CLV by channel, adjust targeting, change onboarding, or stop funding the channel.

Practical Exercise

Build a funnel model with acquisition cost, conversion rate, average order value, gross margin, repeat rate, churn, and CLV.

Real-World Application

Use the model to decide which channel should receive more budget and which should be cut.

Reflection Questions

1. What business signal would you observe first when studying Funnel Analytics, CLV, Churn, and Demand Feedback?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of funnel conversion in this lesson?
2. How does customer lifetime value change the way you interpret business performance?
3. Why does this lesson treat the business as the customer economics system that links acquisition, conversion, retention, and profit.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Convert market behavior into measurable acquisition and retention decisions to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 15: Business Structure, Contracts, and Liability

Phase	Phase 8: Law, Taxation, and Risk Control
Main business science skill	Design legal structure and contracts that reduce avoidable business risk.
Subjects used	Business Law, Taxation, Corporate Finance
Business system being studied	The legal risk control system that defines ownership, obligations, authority, and enforceable business commitments.
Measurable outcome	Create one legal structure and contract risk checklist.

PDF Sources Used

- Business Law / 01_Core_Textbooks/legal-and-ethical-environment-of-business-saylor.pdf
- Business Law / 02_Contracts/american-contract-law-for-a-global-age-cali.pdf
- Business Law / 03_Governance_Legal_Structure/essential-role-of-organizational-law-hansmann-kraakman.pdf
- Taxation / IRS_Pub_583_Starting_a_Business_and_Keeping_Records.pdf
- Taxation / IRS_Form_8832_Entity_Classification_Election.pdf
- Corporate Finance / 32_Jensen_Meckling_Theory_of_the_Firm_Agency_Costs_1976.pdf

Core Concepts

- entity structure
- contract formation
- liability allocation
- agency problems
- recordkeeping
- governance controls

signal → meaning → value → execution Drill

Signal	A recurring customer relationship depends on verbal terms and manual exceptions.
Meaning	Rights, obligations, remedies, and operational responsibilities are unclear.
Value	Ambiguity raises dispute cost, revenue leakage, and liability.
Execution	Document offer, acceptance, performance obligations, payment terms, warranties, and remedies.

Practical Exercise

Draft a business risk map that connects entity choice, contract terms, authority, recordkeeping, and dispute exposure.

Real-World Application

Use the map before signing a vendor, customer, or partner agreement.

Reflection Questions

1. What business signal would you observe first when studying Business Structure, Contracts, and Liability?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of entity structure in this lesson?
2. How does contract formation change the way you interpret business performance?
3. Why does this lesson treat the business as the legal risk control system that defines ownership, obligations, authority, and enforceable business commitments.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design legal structure and contracts that reduce avoidable business risk to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 16: Tax Design, Compliance, and Cash Risk

Phase	Phase 8: Law, Taxation, and Risk Control
Main business science skill	Treat tax as a cash-flow, compliance, and behavior-shaping system.
Subjects used	Taxation, Accounting Information Systems, Financial Accounting
Business system being studied	The tax cash-control system that connects records, timing, deductions, filings, and audit readiness.
Measurable outcome	Create one tax-risk checklist tied to accounting records.

PDF Sources Used

- Taxation / IRS_Pub_334_Tax_Guide_for_Small_Business.pdf
- Taxation / IRS_Pub_538_Accounting_Periods_and_Methods.pdf
- Taxation / IRS_Pub_946_How_to_Depreciate_Property.pdf
- Taxation / IMF_Compliance_Risk_Management_for_Tax_Administration_Leaders.pdf
- Taxation / OECD_Compliance_Risk_Management_Tax_Compliance.pdf
- Accounting Information Systems / 02_Lumen_Financial_Accounting_AIS_and_Bank_Reconciliation.pdf

Core Concepts

- taxable income
- deductions
- depreciation
- accounting method
- tax compliance risk
- bank reconciliation

signal → meaning → value → execution Drill

Signal	Book profit is positive but estimated tax cash is missing.
Meaning	The business measured income but failed to reserve cash for tax obligations.
Value	Tax cash gaps can create penalties, financing stress, and owner withdrawals that weaken the business.
Execution	Build a tax reserve rule, reconcile taxable events monthly, and forecast payments before distributions.

Practical Exercise

Create a tax-risk checklist for a small business covering income, expenses, payroll, depreciation, entity classification, and records.

Real-World Application

Use the checklist to prevent tax surprises from becoming cash crises.

Reflection Questions

1. What business signal would you observe first when studying Tax Design, Compliance, and Cash Risk?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of taxable income in this lesson?
2. How does deductions change the way you interpret business performance?
3. Why does this lesson treat the business as the tax cash-control system that connects records, timing, deductions, filings, and audit readiness.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Treat tax as a cash-flow, compliance, and behavior-shaping system to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 17: Internal Controls, Evidence, and Fraud Prevention

Phase	Phase 9: Auditing and Control Systems
Main business science skill	Design controls that make business information reliable and fraud-resistant.
Subjects used	Auditing, Accounting Information Systems, Business Law
Business system being studied	The verification system that protects the business from unreliable records, weak governance, and misconduct.
Measurable outcome	Draft one internal control checklist with control owner and evidence field.

PDF Sources Used

- Auditing / GAO_Green_Book_Internal_Control_2025.pdf
- Auditing / ACFE_Occupational_Fraud_Report_to_the_Nations_2026.pdf
- Auditing / IIA_Global_Internal_Audit_Standards_2024.pdf
- Auditing / IAASB_Handbook_2025_Volume_1_ISAs_ISQMs.pdf
- Accounting Information Systems / 10_MPRAs_Continuous_Auditing_Technology_Review_2025.pdf
- Business Law / 04_Regulation_Compliance/doj-evaluation-of-corporate-compliance-programs-2024.pdf

Core Concepts

- control environment
- risk assessment
- control activities
- audit evidence
- segregation of duties
- fraud prevention

signal → meaning → value → execution Drill

Signal	One employee can approve vendors, enter invoices, and release payments.
Meaning	Segregation of duties is weak.
Value	Fraud, error, and unauthorized payment risk are concentrated in one role.
Execution	Separate authorization, recording, custody, and review; add exception reporting.

Practical Exercise

Draft an internal control checklist for cash receipts, disbursements, revenue recognition, purchasing, payroll, and system access.

Real-World Application

Use the checklist to decide which controls must be preventive, detective, automated, or manual.

Reflection Questions

1. What business signal would you observe first when studying Internal Controls, Evidence, and Fraud Prevention?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of control environment in this lesson?
2. How does risk assessment change the way you interpret business performance?
3. Why does this lesson treat the business as the verification system that protects the business from unreliable records, weak governance, and misconduct.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design controls that make business information reliable and fraud-resistant to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 18: Data Reliability, Cyber Controls, and Audit Analytics

Phase	Phase 9: Auditing and Control Systems
Main business science skill	Verify whether business data is complete, accurate, timely, and secure enough for decisions.
Subjects used	Auditing, Accounting Information Systems, System Science
Business system being studied	The data assurance system that validates the signals used for accounting, operations, marketing, and finance decisions.
Measurable outcome	Build one data reliability and cyber-control test plan.

PDF Sources Used

- Auditing / GAO_Assessing_Data_Reliability_2019.pdf
- Auditing / GAO_FISCAM_2024.pdf
- Auditing / NIST_SP_800-53r5_Security_Privacy_Controls.pdf
- Auditing / Bayesian_Approach_to_Audit_Evidence_UvA_2026.pdf
- Accounting Information Systems / 11_Informatica_AIS_Big_Data_Mining_Data_Cleaning_Automation.pdf
- System Science / 04_Business_Optimization_and_Decision_Science/algorithms_for_decision_making.pdf

Core Concepts

- data reliability
- access control
- security controls
- audit analytics
- evidence weighting
- decision quality

signal → meaning → value → execution Drill

Signal	Dashboard revenue differs from general ledger revenue.
Meaning	The business has a data definition, timing, completeness, or integration problem.
Value	Bad signals create bad decisions even when the dashboard looks professional.
Execution	Trace source tables, reconcile to ledger, test filters, document definitions, and lock dashboard ownership.

Practical Exercise

Create a data reliability test plan for one dashboard, including source system, field definition, completeness test, accuracy test, and access control.

Real-World Application

Use the test plan before relying on any dashboard for budget, hiring, pricing, or financing decisions.

Reflection Questions

1. What business signal would you observe first when studying Data Reliability, Cyber Controls, and Audit Analytics?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of data reliability in this lesson?
2. How does access control change the way you interpret business performance?

3. Why does this lesson treat the business as the data assurance system that validates the signals used for accounting, operations, marketing, and finance decisions.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Verify whether business data is complete, accurate, timely, and secure enough for decisions to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 19: Transaction Data Architecture and Reporting Standards

Phase	Phase 10: Accounting Information Systems
Main business science skill	Design accounting data flows from events to reports.
Subjects used	Accounting Information Systems, Financial Accounting, Auditing
Business system being studied	The accounting data architecture that captures economic events and produces standardized reports.
Measurable outcome	Create one accounting data flow map from event to report.

PDF Sources Used

- Accounting Information Systems / 08_AISel_REA_Design_Theory_Enterprise_System_Design_2016.pdf
- Accounting Information Systems / 09_Dunn_McCarthy_REA_Intellectual_Heritage_1997.pdf
- Accounting Information Systems / 14_XBRL_US_Financial_Reporting_with_Open_Data_Standards.pdf
- Accounting Information Systems / 17_Springer_Hard_Rules_and_Soft_Rules_for_XBRL_2014.pdf
- Financial Accounting / 21 - REA Triple-Entry Accounting and Blockchain.pdf
- Auditing / GAO_Assessing_Data_Reliability_2019.pdf

Core Concepts

- REA model
- events and resources
- agents
- XBRL

- data standards
- reporting reliability

signal → meaning → value → execution Drill

Signal	Reports require manual spreadsheet adjustments every month.
Meaning	The transaction system is not capturing or classifying events cleanly enough.
Value	Manual reporting increases delay, error, and control risk.
Execution	Redesign event fields, validation rules, account mapping, and report definitions.

Practical Exercise

Design an REA-style event model for sales, purchasing, cash receipts, and cash disbursements.

Real-World Application

Use the model to identify missing fields that prevent reliable financial, operational, or tax reporting.

Reflection Questions

1. What business signal would you observe first when studying Transaction Data Architecture and Reporting Standards?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of REA model in this lesson?
2. How does events and resources change the way you interpret business performance?
3. Why does this lesson treat the business as the accounting data architecture that captures economic events and produces standardized reports.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design accounting data flows from events to reports to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 20: Dashboards, Automation, and Decision Support

Phase	Phase 10: Accounting Information Systems
Main business science skill	Turn transaction data into management dashboards and automated control loops.
Subjects used	Accounting Information Systems, Managerial Accounting, Auditing
Business system being studied	The decision-support system that turns clean data into action thresholds, alerts, and management routines.
Measurable outcome	Build one business dashboard outline with signal, owner, action threshold, and review cadence.

PDF Sources Used

- Accounting Information Systems / 03_Springer_Digital_Dashboard_Use_by_Management_Accountants_2025.pdf
- Accounting Information Systems / 04_Springer_Business_Analytics_and_Management_Control_2025.pdf
- Accounting Information Systems / 12_ABACademies_Business_Intelligence_Activating_AIS_2022.pdf
- Accounting Information Systems / 18_RPA_in_Accounting_Applied_Overview.pdf
- Accounting Information Systems / 19_HighRadius_Bank_Reconciliation_Applied_Overview.pdf
- Managerial Accounting / 20 - Perez-Mendez Machado-Cabezas - Management Information Systems and Corporate Performance.pdf

Core Concepts

- dashboard purpose
- KPI design
- management control
- business intelligence
- robotic process automation
- automated reconciliation

signal → meaning → value → execution Drill

Signal	A KPI is reviewed every week but no decision ever changes.
Meaning	The metric is informational, not operational.
Value	Dashboards waste attention unless each signal has a decision rule.
Execution	Attach every KPI to an owner, threshold, action, and feedback test.

Practical Exercise

Build a dashboard outline with ten KPIs covering demand, margin, cash, throughput, quality, risk, and learning.

Real-World Application

Use the outline to remove vanity metrics and connect each KPI to a decision owner.

Reflection Questions

1. What business signal would you observe first when studying Dashboards, Automation, and Decision Support?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of dashboard purpose in this lesson?
2. How does KPI design change the way you interpret business performance?
3. Why does this lesson treat the business as the decision-support system that turns clean data into action thresholds, alerts, and management routines.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Turn transaction data into management dashboards and automated control loops to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 21: Publishing Product System: IP, Metadata, and Distribution

Phase	Phase 11: Book Publishing Business as Applied Business Science
Main business science skill	Analyze publishing as a product, rights, metadata, and distribution system.
Subjects used	Book Publishing Business, Business Law, Operations Management
Business system being studied	The publishing product system that turns intellectual property into discoverable, saleable, distributed products.
Measurable outcome	Build one publishing product system map.

PDF Sources Used

- Book Publishing Business / 01_IP_Rights/01_WIPO_Understanding_Copyright_and_Related_Rights.pdf
- Book Publishing Business / 01_IP_Rights/04_USCO_Copyright_Basics.pdf
- Book Publishing Business / 01_IP_Rights/07_Authors_Alliance_Publication_Contracts.pdf
- Book Publishing Business / 03_Operations_Metadata/01_EDItEUR_Identifiers_in_ONIX.pdf
- Book Publishing Business / 03_Operations_Metadata/10_EDItEUR_ONIX_Books_Digital_Products.pdf
- Operations Management / 03_KU_Introduction_to_Supply_Chain_and_Operations_Management.pdf

Core Concepts

- copyright
- rights licensing
- publication contracts
- metadata
- identifiers
- distribution workflow

signal → meaning → value → execution Drill

Signal	A title is available but hard to discover in retail systems.
Meaning	The metadata and channel system may be blocking demand from converting.
Value	Poor discoverability lowers revenue without changing product quality.
Execution	Audit identifiers, categories, rights territories, format data, pricing fields, and retailer feed status.

Practical Exercise

Map the lifecycle of one book from rights acquisition to metadata setup, distribution, sales reporting, and royalty accounting.

Real-World Application

Use the map to identify where a publishing business loses value through weak rights, weak metadata, or weak fulfillment.

Reflection Questions

1. What business signal would you observe first when studying Publishing Product System: IP, Metadata, and Distribution?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of copyright in this lesson?
2. How does rights licensing change the way you interpret business performance?
3. Why does this lesson treat the business as the publishing product system that turns intellectual property into discoverable, saleable, distributed products.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Analyze publishing as a product, rights, metadata, and distribution system to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 22: Publishing Profit System: Audience, Pricing, and Open Models

Phase	Phase 11: Book Publishing Business as Applied Business Science
Main business science skill	Use publishing as a case study in demand, pricing, IP economics, and profit design.
Subjects used	Book Publishing Business, Marketing, Cost Accounting, Financial Statement Analysis
Business system being studied	The publishing profit system that connects audience demand, rights, pricing, formats, costs, channels, and cash flow.
Measurable outcome	Build one publishing pricing and profit model.

PDF Sources Used

- Book Publishing Business / 02_Market_Economics/01_WIPO_Global_Publishing_Industry_2022.pdf
- Book Publishing Business / 02_Market_Economics/06_PA_Publishing_in_2025.pdf
- Book Publishing Business / 04_Audience_Demand/01_NEA_Reading_at_Risk.pdf
- Book Publishing Business / 03_Operations_Metadata/04_EDITEUR_Pricing_in_ONIX_Part_1.pdf
- Book Publishing Business / 05_Open_Publishing_Business_Models/04_APC_Revenue_Estimates_Six_Publishers_2019_2023.pdf
- Marketing / 04_Acquisition_Revenue_and_Governance/clv_insights_strategic_marketing.pdf
- Cost Accounting / 18_MDPI_Essential_Factors_Designing_Cost_Accounting_System.pdf

Core Concepts

- publishing market economics
- audience demand
- pricing metadata
- open access models
- customer lifetime value
- cost system design

signal → meaning → value → execution Drill

Signal	Unit sales rise but title-level profit remains weak.
Meaning	Discounts, royalties, format mix, channel fees, or marketing cost may be consuming contribution margin.
Value	Publishing success requires title economics, not just cultural reach.
Execution	Analyze contribution by format and channel, adjust price architecture, and prioritize profitable audience segments.

Practical Exercise

Build a title-level profit model with format mix, list price, discount, unit cost, royalty, marketing cost, and break-even units.

Real-World Application

Use the model to compare traditional sale, subscription, licensing, and open publishing revenue structures.

Reflection Questions

1. What business signal would you observe first when studying Publishing Profit System: Audience, Pricing, and Open Models?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of publishing market economics in this lesson?
2. How does audience demand change the way you interpret business performance?

3. Why does this lesson treat the business as the publishing profit system that connects audience demand, rights, pricing, formats, costs, channels, and cash flow.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Use publishing as a case study in demand, pricing, IP economics, and profit design to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 23: Full Business Diagnosis: From Statements to System Levers

Phase	Phase 12: Integrated Business Science Mastery
Main business science skill	Diagnose a business by connecting financial, operational, market, and risk signals.
Subjects used	System Science, Financial Statement Analysis, Corporate Finance, Operations Management, Marketing, Auditing
Business system being studied	The integrated diagnostic system that converts scattered business data into prioritized improvement levers.
Measurable outcome	Diagnose one business using signal → meaning → value → execution.

PDF Sources Used

- System Science / 02_Dynamics_Feedback_and_Leverage/meadows_leverage_points.pdf
- Financial Statement Analysis / 12_Nissim_Penman_Ratio_Analysis_and_Equity_Valuation_Working_Paper_1999.pdf
- Corporate Finance / 17_Welch_From_Financial_Statements_to_Economic_Cash_Flows_2022.pdf
- Operations Management / 15_Springer_Bottleneck_Identification_Throughput_Serial_Lines_2026.pdf
- Marketing / 02_Market_Understanding_Data_and_Analytics/marketing_mix_modeling_bayesian.pdf
- Auditing / GAO_Green_Book_Internal_Control_2025.pdf

Core Concepts

- multi-signal diagnosis
- cash flow economics
- ratio decomposition
- bottleneck detection

- market response
- control reliability

signal → meaning → value → execution Drill

Signal	Revenue growth, declining cash conversion, longer cycle time, rising churn, and weak approval controls appear together.
Meaning	The business is scaling demand faster than its execution, retention, cash, and control systems can support.
Value	Growth can reduce value when it amplifies system weaknesses.
Execution	Stabilize cash conversion, repair the bottleneck, target better-fit demand, and strengthen controls before scaling further.

Practical Exercise

Diagnose one business using five signals: financial ratio, cash flow, process bottleneck, marketing response, and control weakness.

Real-World Application

Use the diagnosis to rank actions by value impact, evidence strength, execution difficulty, and feedback speed.

Reflection Questions

1. What business signal would you observe first when studying Full Business Diagnosis: From Statements to System Levers?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of multi-signal diagnosis in this lesson?
2. How does cash flow economics change the way you interpret business performance?
3. Why does this lesson treat the business as the integrated diagnostic system that converts scattered business data into prioritized improvement levers.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Diagnose a business by connecting financial, operational, market, and risk signals to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Lesson 24: Build the Business Science Operating System

Phase	Phase 12: Integrated Business Science Mastery
Main business science skill	Design a repeatable operating system for creating, measuring, protecting, and scaling value.
Subjects used	System Science, Accounting Information Systems, Managerial Accounting, Corporate Finance, Business Law, Taxation, Book Publishing Business
Business system being studied	The master operating system that connects strategy, accounting, finance, operations, marketing, risk, data, and execution cadence.
Measurable outcome	Build one Business Science Operating System portfolio.

PDF Sources Used

- System Science / 03_System_Design_and_Engineering/nasa_systems_engineering_handbook.pdf
- Accounting Information Systems / 20_WorldBank_Financial_Management_Information_System_AIS_ERP_Dashboards.pdf
- Managerial Accounting / 09 - CGMA - Global Management Accounting Principles.pdf
- Corporate Finance / 46_Mauboussin_Capital_Allocation_Fundamental_Responsibility_2015.pdf
- Business Law / 04_Regulation_Compliance/doj-evaluation-of-corporate-compliance-programs-2024.pdf
- Taxation / IRS_Pub_583_Starting_a_Business_and_Keeping_Records.pdf
- Book Publishing Business / 05_Open_Publishing_Business_Models/01_Springer_Nature_Future_of_Open_Access_Books.pdf

Core Concepts

- operating system design
- management control
- dashboard governance
- capital allocation
- compliance
- recordkeeping
- applied business model synthesis

signal → meaning → value → execution Drill

Signal	The business has many metrics but no shared decision rhythm.
Meaning	Information exists, but the operating system does not convert it into disciplined action.
Value	Value creation requires repeated execution, not isolated analysis.
Execution	Install weekly, monthly, quarterly, and annual review loops with owners, thresholds, actions, and evidence standards.

Practical Exercise

Assemble the final project components into one operating manual with decision rules, dashboards, controls, and feedback loops.

Real-World Application

Use the operating system as a repeatable template for analyzing, improving, building, or scaling any business.

Reflection Questions

1. What business signal would you observe first when studying Build the Business Science Operating System?
2. What assumption in this lesson would you test before trusting the conclusion?
3. What action would you take if the signal contradicted your original business belief?

Comprehension Questions

1. What is the main purpose of operating system design in this lesson?

2. How does management control change the way you interpret business performance?
3. Why does this lesson treat the business as the master operating system that connects strategy, accounting, finance, operations, marketing, risk, data, and execution cadence.?

Diagnostic Questions

1. Which signal would show that this system is underperforming?
2. Which measurement could be misleading if the source data is incomplete?
3. What constraint, risk, or feedback delay would you inspect before recommending action?

Application Questions

1. Apply Design a repeatable operating system for creating, measuring, protecting, and scaling value to a real business and state the next action.
2. What data would you collect this week to test the lesson's core idea?
3. What simple model, checklist, map, or dashboard would you build from this lesson?

Systems-Thinking Questions

1. Where are the inputs, transformations, outputs, feedback loops, and constraints?
2. How could improving one part of the system damage another part?
3. What leverage point would create the largest improvement with the least waste?

Mastery Questions

1. Can you explain the lesson using only signal → meaning → value → execution?
2. Can you defend your diagnosis with accounting, operational, market, or risk evidence?
3. Can you turn the diagnosis into a measurable execution rule with a review cadence?

Weekly Schedule

Cadence: one lesson per week. Minimum weekly output: one artifact, one diagnosis, one execution rule, and one feedback metric. Spend 60 percent of study time on reading and extraction, 25 percent on the practical artifact, and 15 percent on reflection and review.

Week	Lesson	Primary Output	Feedback Loop
1	Business Science Operating Logic	Build one business system map using signal → meaning → value → execution.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
2	Business Model as Value Conversion Engine	Create one business model map with assumptions, signals, and decision rules.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
3	Transaction Flow and Financial Statement Architecture	Trace ten transactions through one accounting system map.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
4	Accounts as Signals of Business Behavior	Create one account-signal dictionary for a target business.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
5	Ratio Diagnostics for Business Health	Calculate five financial ratios and convert each into one business action hypothesis.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
6	Earnings Quality, Fraud Signals, and Failure Risk	Build one financial risk red-flag checklist.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
7	Cost Behavior, Margins, and Pricing Floors	Create one cost behavior and pricing model.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
8	Budgeting, Variance, and Internal Performance Control	Build one budget variance analysis and corrective action log.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
9	Time Value, NPV, and Investment Decisions	Build one capital budgeting model with sensitivity analysis.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
10	Risk, Capital Structure, and Long-Term Value	Draft one capital allocation and financing decision memo.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
11	Flow, Throughput, and Bottleneck Diagnosis	Identify one operational bottleneck and one throughput improvement action.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
12	Lean Quality, Capacity, and Scalable Execution	Create one operations improvement plan with before-and-after metrics.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
13	Customer Behavior, Positioning, and Offer Design	Create one offer and positioning model for a target customer segment.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
14	Funnel Analytics, CLV, Churn, and Demand Feedback	Analyze one marketing funnel and calculate customer lifetime value.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
15	Business Structure, Contracts, and Liability	Create one legal structure and contract risk checklist.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
16	Tax Design, Compliance, and Cash Risk	Create one tax-risk checklist tied to accounting records.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
17	Internal Controls, Evidence, and Fraud Prevention	Draft one internal control checklist with control owner and evidence field.	Review the artifact using signal → meaning → value → execution and revise one decision rule.

Week	Lesson	Primary Output	Feedback Loop
18	Data Reliability, Cyber Controls, and Audit Analytics	Build one data reliability and cyber-control test plan.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
19	Transaction Data Architecture and Reporting Standards	Create one accounting data flow map from event to report.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
20	Dashboards, Automation, and Decision Support	Build one business dashboard outline with signal, owner, action threshold, and review cadence.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
21	Publishing Product System: IP, Metadata, and Distribution	Build one publishing product system map.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
22	Publishing Profit System: Audience, Pricing, and Open Models	Build one publishing pricing and profit model.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
23	Full Business Diagnosis: From Statements to System Levers	Diagnose one business using signal → meaning → value → execution.	Review the artifact using signal → meaning → value → execution and revise one decision rule.
24	Build the Business Science Operating System	Build one Business Science Operating System portfolio.	Review the artifact using signal → meaning → value → execution and revise one decision rule.

Business Science Skill Tracker

Skill	System Function	Evidence of Mastery	Measurement
System mapping	Organizing logic	Business map with inputs, process, outputs, constraints, feedback	One map per business analyzed
Accounting measurement	Business signal capture	Transactions traced to statements	Ten transactions classified correctly
Financial diagnosis	Performance interpretation	Ratio and cash-flow diagnosis	Five ratios plus one action hypothesis each
Cost modeling	Margin control	Cost behavior and contribution model	Break-even and target-profit units calculated
Capital allocation	Value creation	NPV and risk memo	Project decision defended with cash-flow model
Operations improvement	Execution system	Bottleneck and value stream analysis	Cycle time, WIP, throughput, and defect metrics
Demand creation	Market pull	Offer, funnel, CLV, and churn model	Channel-level conversion and CLV calculated
Legal and tax control	Risk reduction	Entity, contract, record, and tax checklist	Risk owner and evidence field assigned
Audit control	Information reliability	Internal control and data reliability test plan	Control design matched to risk
AIS design	Reporting and automation	Data flow, dashboard, and automation outline	Each KPI has owner, threshold, and action
Integrated diagnosis	Mastery	Full business science operating system	Final project passes rubric

Financial Analysis Exercise Bank

- Build a common-size income statement and identify the largest margin pressure.
- Calculate current ratio, quick ratio, debt-to-equity, gross margin, operating margin, ROA, ROE, asset turnover, inventory days, and days sales outstanding.
- Convert accounting earnings into operating cash flow hypotheses.
- Compare revenue growth to receivables growth and identify collection risk.
- Run a DuPont-style profitability decomposition.
- Build a three-scenario cash-flow forecast.
- Estimate working capital needs for a growing business.
- Compare book profit, taxable income, and cash available for distribution.
- Create an earnings-quality red-flag matrix.
- Build an Altman-style distress signal checklist.
- Compare two businesses using ratios and decide which system is healthier.
- Write a one-page financial diagnosis using signal → meaning → value → execution.

Operations Improvement Exercise Bank

- Draw a process map with inputs, transformation steps, outputs, owners, controls, and customers.
- Measure cycle time, wait time, rework, and throughput for one workflow.
- Apply Little's Law to estimate WIP, cycle time, or throughput.
- Identify the constraint and write one exploit, subordinate, elevate, and protect action.

- Build a value stream map with current-state and future-state metrics.
- Estimate the cost of poor quality from rework, scrap, returns, and delay.
- Design a capacity plan for a demand surge.
- Build a process dashboard with leading and lagging indicators.
- Use material flow costing to expose waste that standard costs hide.
- Create a root-cause tree for one recurring operational defect.
- Write a standard work checklist for a critical process.
- Run one improvement experiment and define the feedback signal before acting.

Marketing and Demand Exercise Bank

- Define a target segment by problem, urgency, willingness to pay, and channel behavior.
- Create a positioning statement with alternative, difference, proof, and risk reversal.
- Map one funnel from impression to retained customer.
- Calculate acquisition cost, conversion rate, average order value, gross margin, repeat rate, churn, and CLV.
- Segment customers using recency, frequency, and monetary value.
- Compare two channels by contribution margin after acquisition cost.
- Build a churn diagnosis by cohort.
- Design an offer test with hypothesis, variable, metric, threshold, and decision rule.
- Audit brand proof: testimonials, guarantees, demonstrations, samples, and authority signals.
- Map price sensitivity by segment and value metric.
- Create a retention loop with onboarding, usage trigger, feedback request, and next offer.
- Write a demand diagnosis using signal → meaning → value → execution.

Legal, Tax, and Audit Risk Checklist

- Entity structure matches ownership, liability, tax, and financing needs.
- Authority to sign contracts is documented.
- Standard customer, vendor, employee, and partner agreements exist.
- Offer, acceptance, consideration, performance duties, warranties, remedies, and termination terms are clear.
- Intellectual property ownership and licensing terms are documented.
- Customer claims and advertising promises are supportable.
- Revenue recognition aligns with contract obligations and delivery evidence.
- Tax entity classification is documented.
- Accounting method, tax year, and recordkeeping policy are documented.
- Payroll, contractor, sales, income, and excise tax obligations are mapped.
- Estimated tax cash reserve rule exists.
- Depreciation, basis, and asset records are maintained.
- Bank accounts are reconciled monthly.
- Segregation of duties exists for cash, purchasing, payroll, and system access.
- Vendor approval, invoice approval, and payment release are separated.
- Dashboard definitions reconcile to accounting records where relevant.
- Access rights are reviewed periodically.
- Exception reports are reviewed by an owner.
- Audit trail exists for material transactions.
- Fraud red flags have escalation rules.
- Cybersecurity controls protect financial and customer data.
- Management reviews control failures and remediation status.
- Compliance training and reporting channels exist where needed.
- Key risks have owners, evidence, threshold, and review cadence.
- Every risk signal is translated into meaning, value, and execution.

Business Systems Diagnostic Template

Field	Prompt
System boundary	What business system is being diagnosed: demand, accounting, finance, operations, risk, data, or integrated business?
Objective	What value should this system create or protect?
Signal	What data, behavior, financial result, operational issue, customer action, legal risk, tax issue, or market feedback is observed?
Meaning	What does the signal reveal about system condition, constraint, risk, or opportunity?
Value	Why does it matter for profit, cash flow, risk control, growth, efficiency, or survival?
Execution	What action should be taken, by whom, by when, and with what constraint?
Evidence quality	How reliable is the data source, definition, timing, completeness, and control trail?
Feedback metric	Which metric will prove whether the execution improved the system?
Review cadence	Daily, weekly, monthly, quarterly, or annual?
Learning loop	What changed in the system model after the feedback came in?

Mastery Assessment

- Closed-book framework test: define every major business system and its primary signals.
- Accounting test: trace transactions into statements and interpret account behavior.
- Financial analysis test: calculate and interpret ratios, cash-flow signals, and distress indicators.
- Cost and margin test: build a contribution model and pricing decision.
- Finance test: evaluate one investment with NPV, risk, and capital constraints.
- Operations test: map a process, identify the bottleneck, and design an improvement experiment.
- Marketing test: build a funnel and CLV model and defend channel allocation.
- Risk test: produce legal, tax, audit, and control checklists for one business.
- AIS test: design data flow, dashboard definitions, and control evidence.
- Integration test: diagnose one business using signal → meaning → value → execution across all systems.

Final Business Science Project

The Business Science Operating System

Purpose: prove mastery by building one repeatable operating system that explains how a business creates, measures, protects, and scales value.

- One business model map.
- One accounting system map.
- One financial statement analysis.
- One cost and margin analysis.
- One operations improvement plan.
- One marketing and demand creation plan.
- One legal and tax risk checklist.
- One audit and internal control checklist.
- One accounting information system design.
- One business science synthesis essay explaining how the business creates, measures, protects, and scales value.

Component	Mastery Standard
Business model	Shows customer, offer, capability, revenue, cost, capital, and feedback logic.
Accounting	Connects transactions, records, statements, controls, and decision signals.
Finance	Defends capital allocation with cash flow, risk, return, and long-term value.
Operations	Identifies bottleneck, throughput, quality, capacity, and improvement experiment.
Marketing	Links segment, offer, channel, funnel, retention, CLV, and market feedback.
Risk	Maps legal, tax, audit, compliance, fraud, and control exposures with owners.
AIS	Designs data capture, reporting standards, dashboard metrics, automation, and evidence trail.
Synthesis	Uses signal → meaning → value → execution to connect every subsystem into one operating rhythm.

Long-Term Practice System

Cadence	Practice	Output
Daily	Observe one business signal from accounting, customers, operations, or risk.	Signal log entry
Weekly	Convert three signals into meaning, value, and execution.	One action rule and one feedback metric
Monthly	Run a mini financial statement and operations diagnosis.	Ratio sheet, bottleneck note, cash-risk note
Quarterly	Review strategy, capital allocation, marketing economics, controls, and tax exposure.	Quarterly business science memo
Annual	Rebuild the operating system map and compare actual value creation to the model.	Annual business science operating review